

Luke Liang

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EDUCATION

Columbia University, School of Engineering and Applied Science

M.S. in Computer Science, Machine Learning Track, GPA 3.66

New York, NY

Expected Dec 2026

Miami University, College of Engineering and Computing

B.S. in Computer Science, B.S. in Data Science & Statistics, Minor in Math, GPA 3.86

Oxford, OH

Aug 2021 - May 2024

- Astronaut Scholarship (national), Summa Cum Laude, Honors College, ML training on Oak Ridge National HPC

EXPERIENCE

Columbia University

Research Assistant

Sept 2025 – Present

New York, NY

- Automated a bioinformatics pipeline for high-throughput viral genomic sequencing, successfully processing 400+ patient samples and 5+ TB of data
- Transformed complex genomic outputs into actionable clinical insights, developing automated visualization scripts to generate diagnostic charts and variant summary dashboards
- Optimized file I/O and storage handling for raw sequencing runs, accelerating processing speeds when managing multi-terabyte viral dataset loads.

VISA

Software Engineering Intern

May 2026 – Aug 2026

Austin, TX

- Architected an AI-driven Container Vulnerability Remediation Agent leveraging Claude via MCP, cutting manual time-to-patch by 40% across 1,500+ supported container images
- Automated end-to-end pipelines by provisioning backend database, API integration with production servers, GitHub integrations to parse repository manifest files, submit patch PRs, and triggers for SMTP status alerts
- Deployed a conversational AI assistant in MS Teams using Copilot Studio, enabling developers to query container vulnerability states and trigger automated remediation workflows directly from chat
- Engineered safety guardrails within the Model Context Protocol (MCP) framework to validate AI-generated patch recommendations and prevent unexpected breaking changes before PR creation.
- Designed relational database schemas and indexing strategies to maintain immutable audit logs of container vulnerability histories, patch statuses, and compliance metrics.

Miami University

Research Assistant

Aug 2021 – Oct 2024

Oxford, OH

- Engineered a high-performance simulation platform for CDC researchers to evaluate youth suicide prevention policies, translated real-world public health phenomena into scalable quantitative models in close collaboration with domain experts, and built custom visualization tools for multi-variable scenario.
- Accelerated core simulation algorithms using Numba, NumPy, and multithreading on high-performance compute nodes, enabling real-time execution of multi-agent population models.
- Built interactive state-management logic into the CDC platform, enabling researchers to dynamically adjust parameters and render real-time overlay charts for policy comparison.
- Pioneered anomaly detection pipelines achieving up to 95% accuracy in identifying erroneous network simulations by developing novel graph-to-image encodings, few-shot learning methods, and compression techniques.
- Benchmarked graph-to-image classification architectures against traditional graph feature extraction methods to demonstrate the viability of computer vision in network science validation.
- Built a stochastic HIV viral simulation in Python and NumPy, integrating Stanford's Drug Resistance Database and clinical mutation statistics to model cell-to-cell infection dynamics under 1-, 2-, and 3-drug treatment regimens.
- Executed Monte Carlo stochastic sampling to model cell-to-cell viral replication dynamics, mutation odds, and treatment failure rates across single- and multi-drug regimes.

JPMorgan Chase & Co.

Software Engineering Intern

Jun 2022 – Aug 2022

Chicago, IL

- Migrated legacy AngularJS applications to React and managed infrastructure deployments, enhancing front-end performance, component maintainability, and delivery reliability across internal systems.

- Refactored legacy monolithic controllers into modular React functional components and custom hooks, significantly reducing bundle sizes and front-end rendering latency.
- Authored unit and integration test suites to verify UI behavior, ensure component accessibility, and maintain application stability throughout the migration process.
- Configured automated build and deployment pipelines to streamline staging releases and lower risk across internal infrastructure environments.
- Won 1st Place at the JPMorgan Code for Good Hackathon (2022), collaborating with a team of interns to design and deploy a full-stack web application for a non-profit partner.

PUBLICATIONS

- **L. Liang**, et al. An Artificial Intelligence Approach to Support Adolescent Suicide Prevention Initiatives in the United States. Artificial Intelligence in Medicine (2026)
- **L. Liang**, H. Phan, P.J. Giabbanelli. Experimental Evaluation of a Machine Learning Approach to Improve the Reproducibility of Network Simulations. Simulation (2024)
- R. Schuerkamp, **L. Liang**, K.L. Rice, P.J. Giabbanelli. Simulation Models for Suicide Prevention: A Survey of the State-of-the-Art." Computers, Vol. 12, No. 7, Art. 132 (2023)
- M. Nguyen, D. Vu, A. Vo, **L. Liang**, P.J. Giabbanelli. Preserving Simulation Insight While Removing Data: Verification of Compressed Simulation Traces via Machine Learning. Proceedings of the 2023 Annual Modeling and Simulation Conference (ANNSIM), pp. 345–356 (2023)
- M.K. Wozniak, **L. Liang**, H. Phan, P.J. Giabbanelli. A New Application of Machine Learning: Detecting Errors in Network Simulations. Proceedings of the 2022 Winter Simulation Conference (WSC), pp. 653–664 (2022)

PRESENTATIONS (presenting author first)

- Oral Presentation. **L. Liang**, M. Nguyen, D. Vu, A. Vo, P. J. Giabbanelli. Preserving simulation insight while removing data: verification of compressed simulation traces via machine learning. ANNSIM conference. May 2023
- Oral Presentation. **L. Liang**, M. K. Wozniak, H. Phan, P. J. Giabbanelli. A new application of machine learning: detecting errors in network simulations. WSC online. Dec 2022.
- Poster Presentation. **L. Liang**, M. K. Wozniak, H. Phan, P. J. Giabbanelli. A new application of machine learning: detecting errors in network simulations. COF Scholar showcase online. Feb 2022.

SKILLS

Languages: Python, Java, C/C++, R, SAS, SQL, JavaScript/TypeScript, HTML/CSS

AI/ML & Data: PyTorch, scikit-learn, pandas, NumPy, Tensorflow, MATLAB

AI, Web, Cloud & Databases: React, Node.js, mySQL, Supabase, AWS Bedrock/Lambda/S3, Vercel

Developer & AI Tools: Claude Code, OpenAI Codex, Git/GitHub, Linux/Unix